SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned checkpoint — 19 July 2026

Authority. Corrected French lines 2874–2886; original printed pp. 96–97; physical source-PDF p. 84; recomposed running p. 76. Structural blanks 2875, 2877, and 2881 are included; blank line 2887 is excluded. The printed-page-97 token occurs within French line 2886 immediately after *vérifiée*. The corrected branch restores "of page 74": this is recomposed running p. 74 (original printed p. 93; physical source-PDF p. 82), not printed p. 74. French line 2888 begins Proposition 3.2 and is the exact continuation cursor. The compiled French reader is generated from the same corrected edition and is page/layout evidence, not independent corroboration. The English target's calligraphic $\mathcal{F}, \mathcal{G}$ are a provisional, explicitly adjudicated normalization from the corrected source's upright operator F, G; they are not literal source-glyph preservation.

3. Applications

From these results one deduces a coherence condition for the higher direct images of a coherent sheaf under a *morphism that is not proper.*

Theorem 3.1.—Let $f\colon X \to Y$ be a morphism of preschemes. Suppose that Y is *locally Noetherian* and that f is *proper.* Suppose that X is *locally embeddable in a regular prescheme.* Let $n \in \mathbb{Z}$. Let U be an open subset of X, and let $\mathcal{F}$ be a *coherent* $\mathcal{O}_U$-module. Suppose that, for every $x \in U$ such that

$$\operatorname{codim}(\overline{\{x\}} \cap (X \setminus U), \overline{\{x\}}) = 1,$$

one has $\operatorname{prof} \mathcal{F}_x \geq n$. *Then* the $\mathcal{O}_Y$-modules

$$R^p(f \circ g)_*(\mathcal{F})$$

are coherent for $p < n$, where g is the canonical immersion of U into X.

Indeed, there is a Leray spectral sequence whose abutment is $R^*(f \circ g)_*(\mathcal{F})$ and whose initial term is given by

$$E_2^{p,q} = R^p f_*(R^q g_*(\mathcal{F})).$$

Moreover, there exists a *coherent* $\mathcal{O}_X$-module $\mathcal{G}$ such that $\mathcal{G}|_U \simeq \mathcal{F}$ (EGA I, 9.4.3). It then follows from the preceding section that condition (iv) on p. 74 is satisfied [French printed p. 97], i.e. that $R^q g_*(\mathcal{G}|_U)$ is coherent for $q < n$. Applying the finiteness theorem of EGA III, 3.2.1 to f and the sheaves $R^q g_*(\mathcal{F})$, one finds that $E_2^{p,q}$ is coherent for $q < n$, whence the conclusion.